

Guide to Install MATLAB and Simulink for Ubuntu

Specific guide for WingLoop use, March 2026

0.0) Introduction

This guide was written by Leonardo AVONI (avonileonardo@gmail.com) to ease the install procedure of MATLAB and Simulink, to be used within the WingLoop framework. Have fun!

1.0) MATLAB Setup

1.1) Download and Install MATLAB

To access MATLAB zip file:

<https://fr.mathworks.com/help/install/ug/install-products-with-internet-connection.html>

Basically:

1. Download the Matlab ubuntu zip
2. Unzip it using the following commands (and move to the folder):

```
unzip matlab_R2025b_Linux.zip -d ./matlab_R2025b_Linux
cd ./matlab_R2025b_Linux
```

3. Launch the installation procedure

```
xhost +SI:localuser:root
sudo -H ./install
xhost -SI:localuser:root

# Alias for Matlab R2025b
alias matlab='/usr/local/MATLAB/R2025b/bin/matlab'
```

For the WingLoop setup, it is convenient to install now the following packages:

- MATLAB Compiler
- Simulink Compiler
- FMU Builder for Simulink

If issues appear for package installation, refer to the dedicated section below.

When done with the installation, you may need to check if MATLAB already runs from the terminal just by typing `matlab`. If not, you can first check if it runs by typing the whole MATLAB path:

```
/usr/local/MATLAB/R2025b/bin/matlab
```

If `matlab` opens (the installation worked), it means the MATLAB shortcut is not there. You will have to modify the `.bashrc` file as follows (the following line will open the `.bashrc` file located in the Home directory, using the nano software). You can use the same file if you navigate to the Home directory, and open the (hidden) `.bashrc` file with another software:

```
nano ~/.bashrc
```

Once there, paste this into the `.bashrc` file:

```
# Alias for Matlab R2025b
alias matlab='/usr/local/MATLAB/R2025b/bin/matlab'
```

When done: `Ctrl+O`, `Enter` (saves the file and confirms the name), `Ctrl+X` (closes the nano editor)

Then, to actuate the changes made, either restart the computer (long), or type the following line:

```
source ~/.bashrc
```

You now should be able to run `matlab` from the terminal by typing just `matlab`

1.2) Create a MATLAB Desktop Entry

Open a terminal and run the following command. This creates and opens with nano an empty `matlab.desktop` file.

```
sudo nano /usr/share/applications/matlab.desktop
```

Paste this into the `matlab.desktop` file:

```
[Desktop Entry]
```

```
Type=Application
Name=MATLAB R2025b
Icon=/usr/local/MATLAB/R2025b/extern/icons/matlab.png
Exec=/usr/local/bin/matlab_open.sh %f
Comment=MATLAB - Numerical Computing Environment
Categories=Development;Science;Math;
Terminal=false
StartupNotify=true
MimeType=text/x-matlab;
```

When done: Ctrl+O, Enter (saves the file and confirms the name), Ctrl+X (closes the nano editor)

Before continuing, you need to make the `matlab.desktop` entry executable:

```
sudo chmod +x /usr/share/applications/matlab.desktop
```

1.3) Create a MATLAB Launcher File

Type the following (mkdir the bin directory if it does not exist). This creates an empty `matlab_open.sh` file

```
nano /usr/local/bin/matlab_open.sh
```

Paste the following in that file

```
#!/bin/bash

FILE="$1"

if [ -z "$FILE" ]; then
    /usr/local/MATLAB/R2025b/bin/matlab -desktop
else
    DIR="$(dirname "$FILE")"
    BASE="$(basename "$FILE")"
    /usr/local/MATLAB/R2025b/bin/matlab -desktop -r "cd('$DIR');"
    edit('$BASE');"
fi
```

When done: Ctrl+O, Enter (saves the file and confirms the name), Ctrl+X (closes the nano editor)

Before continuing, you need to make the `matlab_open.sh` file executable:

```
chmod +x /usr/local/bin/matlab_open.sh
```

When done, you need to refresh the applications database. To do so, either type the following:

```
sudo update-desktop-database
```

Or simply log out and log back in.

Now you should see **MATLAB R2025b** in your Applications menu with an icon.

1.3) Create a MATLAB Icon

The installer normally does not take care of the MATLAB icon, hence

1. Create (or find somewhere) an icon called `matlab.png`
2. Create (if not already there) the icons folder in the MATLAB folder:

```
sudo mkdir /usr/local/MATLAB/R2025b/extern/icons/
```

3. Move the `matlab.png` icon to that folder:

```
sudo mv matlab.png /usr/local/MATLAB/R2025b/extern/icons/
```

1.3) Set-Up the MATLAB Python Engine

NOTE: Only proceed with this part if you already set-up a proper Python environment.

To install the MATLAB-Python API; Matlab specifies some installation procedures:

- https://fr.mathworks.com/help/matlab/matlab_external/install-the-matlab-engine-for-python.html

If you follow those + chatgpt and try to install it using the default install file, it will work (maybe) but there may be issues. At the moment, it seems that the link is available for the python 3.12.x versions; 3.13.x do not work; so make sure you install it in an environment that runs 3.12.x

We have found the following command to work (installs directly from online the latest version). Moreover, it's more straight-forward. The only problem is that it still (at the moment, March 2026) only works for Python 3.12.x ("Python 3.9, 3.10, 3.11, or 3.12 ") but it's ok

```
python -m pip install matlabengine
```

1.4) Removing old MATLAB or Python Links

When typing “matlab” or “simulink” in the terminal, it is possible that Ubuntu does not understand where to look for the app. It is possible that the following files exist in the ~/.local/share/applications/ folder:

```
mw-matlabconnector.desktop  
mw-matlab.desktop  
mw-simulink.desktop
```

If so, remove them:

```
rm ~/.local/share/applications/mw-matlabconnector.desktop  
rm ~/.local/share/applications/mw-matlab.desktop  
rm ~/.local/share/applications/mw-simulink.desktop
```

2.0) Simulink Setup

2.1) Create a Simulink Desktop Entry

Open a terminal and run the following command. This creates and opens with nano an empty simulink-r2025b.desktop file.

```
sudo nano /usr/share/applications/simulink-r2025b.desktop
```

Paste this into the simulink-r2025b.desktop file:

```
[Desktop Entry]  
Type=Application  
Name=Simulink R2025b  
Comment=Simulink - Model-Based Design Environment  
Exec=/usr/local/bin/simulink_open.sh %f  
Icon=/usr/local/MATLAB/R2025b/extern/icons/simulink.png  
Terminal=false  
StartupNotify=true  
Categories=Development;Science;Math;  
MimeType=application/x-simulink;
```

When done: Ctrl+O, Enter (saves the file and confirms the name), Ctrl+X (closes the nano editor)

Before continuing, you need to make the `simulink-r2025b.desktop` entry executable:

```
sudo chmod +x /usr/share/applications/simulink-r2025b.desktop
```

2.2) Create a Simulink Launcher File

Type the following (mkdir the bin directory if it does not exist). This creates an empty `simulink_open.sh` file

```
nano /usr/local/bin/simulink_open.sh
```

Paste the following in that file

```
#!/bin/bash

FILE="$1"

if [ -z "$FILE" ]; then
    # No file provided → just open Simulink
    /usr/local/MATLAB/R2025b/bin/matlab -desktop -r "simulink"
else
    # File provided → extract directory and base name
    DIR="$(dirname "$FILE")"
    BASE="$(basename "$FILE")"
    /usr/local/MATLAB/R2025b/bin/matlab -desktop -r "cd('$DIR');
open_system('$BASE');"
fi
```

When done: Ctrl+O, Enter (saves the file and confirms the name), Ctrl+X (closes the nano editor)

Before continuing, you need to make the `matlab_open.sh` file executable:

```
chmod +x /usr/local/bin/matlab_open.sh
```

When done, you need to refresh the applications database. To do so, either type the following:

```
sudo update-desktop-database
```

Or simply log out and log back in.

Now you should see **Simulink R2025b** in your Applications menu with an icon.

2.3) Create a MATLAB Icon

Same as for MATLAB, create a dedicated icon, then place it to the proper folder as follows:

```
sudo mv simulink.png /usr/local/MATLAB/R2025b/extern/icons/
```

2.0) Installing Packages

For WingLoop use, the following MATLAB Add-Ons need to be installed:

- MATLAB Compiler
- Simulink Compiler
- FMU Builder for Simulink

It is possible that MATLAB creates issues when attempting to install MathWorks Add-Ons. If so, for the case of the **Simulink Compiler** and **Matlab Compiler** packages, it is possible that one needs to activate the C++ libraries:

```
mex -setup  
mex -setup cpp
```

It is also possible that matlab blocks the package installation if it is not opened as sudo. If this event happens, you have two options:

- Either you run the installer again, selecting the appropriate packages (same commands as above, copied here for sake of use)

```
xhost +SI:localuser:root  
sudo -H ./install  
xhost -SI:localuser:root
```

- Or open MATLAB as sudo:

```
sudo /usr/local/MATLAB/R2025b/bin/matlab
```